

Abstract

Abstract I

Abstract II

No prior automated system tracks hypothesis-level evidence mapping across the full Active Inference and Free Energy Principle (FEP) literature. Manual synthesis cannot keep pace with a field that has grown at a compound annual rate of 20.36% across 2000–2026 (empirical publication span 2005–2026), and the FEP's theoretical generality has invited falsifiability critiques that only hypothesis-specific triage can address. Building on pioneering systematic manual annotation paired with ontology-based analysis at the scale of hundreds of papers, we present a computational meta-analysis framework that automates and scales this approach. The pipeline retrieves literature from arXiv, Semantic Scholar, and OpenAlex, deduplicating $N = 817$ papers (inclusion from 2000 onward) via a canonical identifier hierarchy (DOI > arXiv ID > Semantic Scholar ID > OpenAlex ID). It classifies papers into a three-tier taxonomy spanning eight categories: A (Core Theory), B (Tools & Translation), and C (Application Domains). An LLM-powered extraction system evaluates each abstract against eight core hypotheses, separating (1) explicit source claims, (2) whether the abstract supplies evidence, and (3) pipeline hypothesis